COURSE OUTCOME 1

PROGRAM NO:1

DATE: 14/09/26

LOGIN FORM

AIM: Design a Login Form with uscrname and password using Lincar Layout and toast valid credentials.

MainActivity.java

```
package com.example.program1;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
   EditText e_name, e_pwd;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);


      e_name = findViewById(R.id.ed_username);
      e_pwd = findViewById(R.id.ed_passwd);
   }
   public void onClick(View view) {
      String s1 = e_name.getText().toString();
      String s2 = e_pwd.getText().toString();
      if (s1.equals("MCA") && s2.equals("mca")) {
         Toast.makeText(
               MainActivity.this,
               "LOGIN SUCCESSFUL",
               Toast.LENGTH_SHORT
         ).show();
```

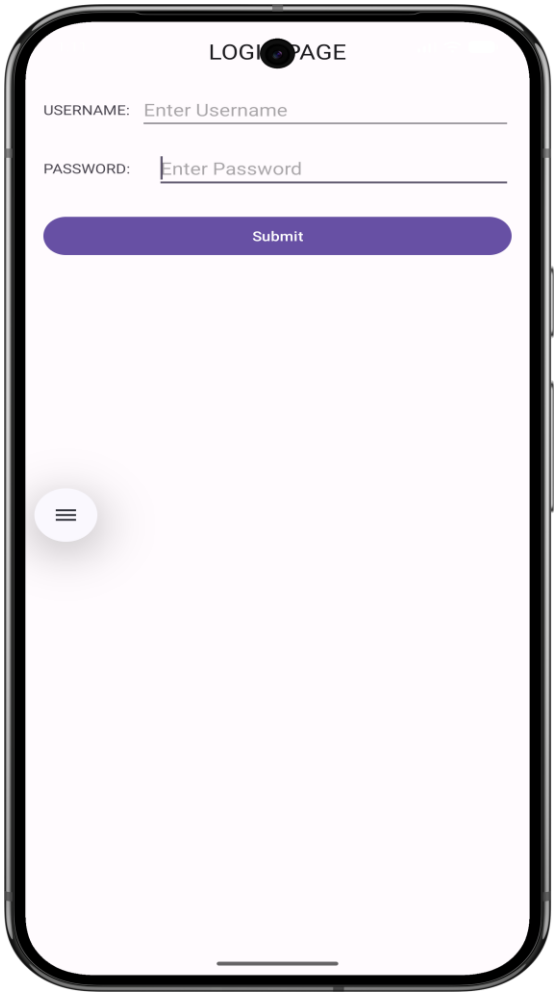
```
        } else {
            Toast.makeText(
                    MainActivity.this,
                    "INVALID USERNAME OR PASSWORD",
                    Toast.LENGTH_SHORT
            ).show();
        }
    }
}
```

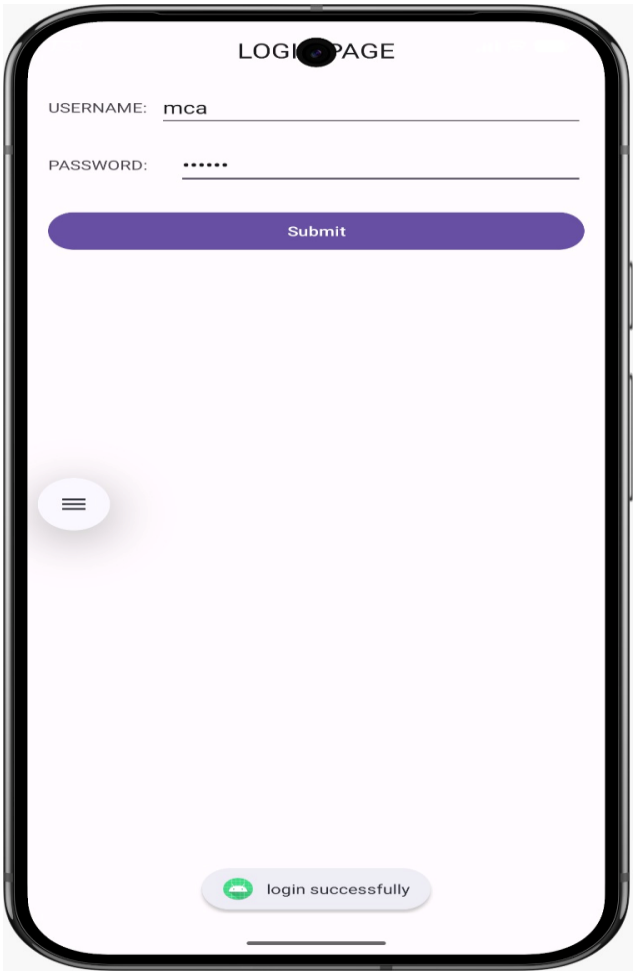
activity.xml:

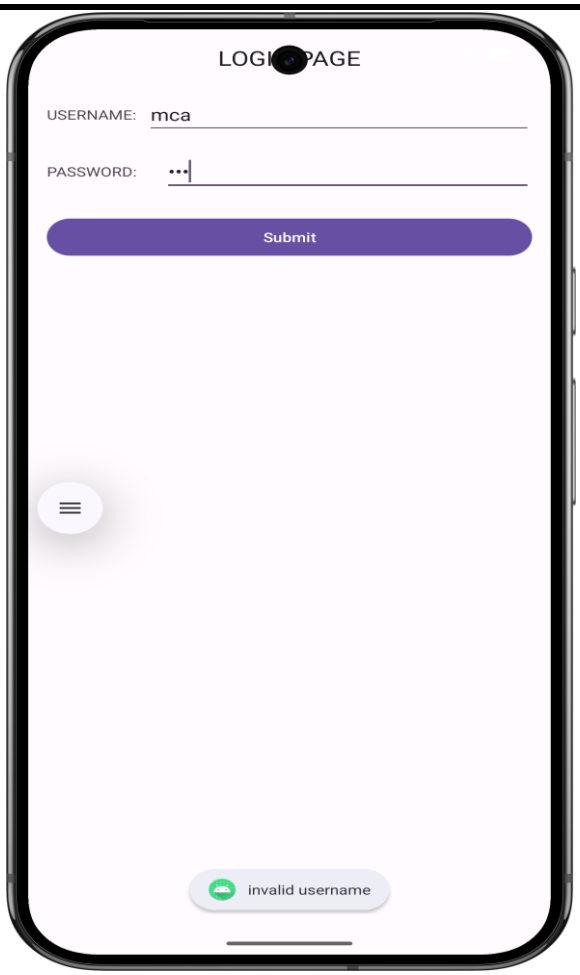
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    tools:context=".MainActivity">
    <TextView
        android:id="@+id/textView5"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login Page"
        android:textSize="24sp" />
    <EditText
        android:id="@+id/ed_username"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"
        android:inputType="text" />
    <EditText
        android:id="@+id/ed_passwd"
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword" />
    <Button
        android:id="@+id/btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:onClick="onClick"
        android:text="Login" />
</LinearLayout>
```

output:







RESULT :

The mobile application user interface was successfully designed and developed using basic building blocks and UI components. The application structure was implemented and tested on the Emulator, and the expected output screens were obtained.

PROGRAM NO:2

DATE: 14/09/26

ACTIVITY LIFECYCLE

AIM: Write a program that demonstrates Activity Lifecycle.

MainActivity.java

mainactivity.java:

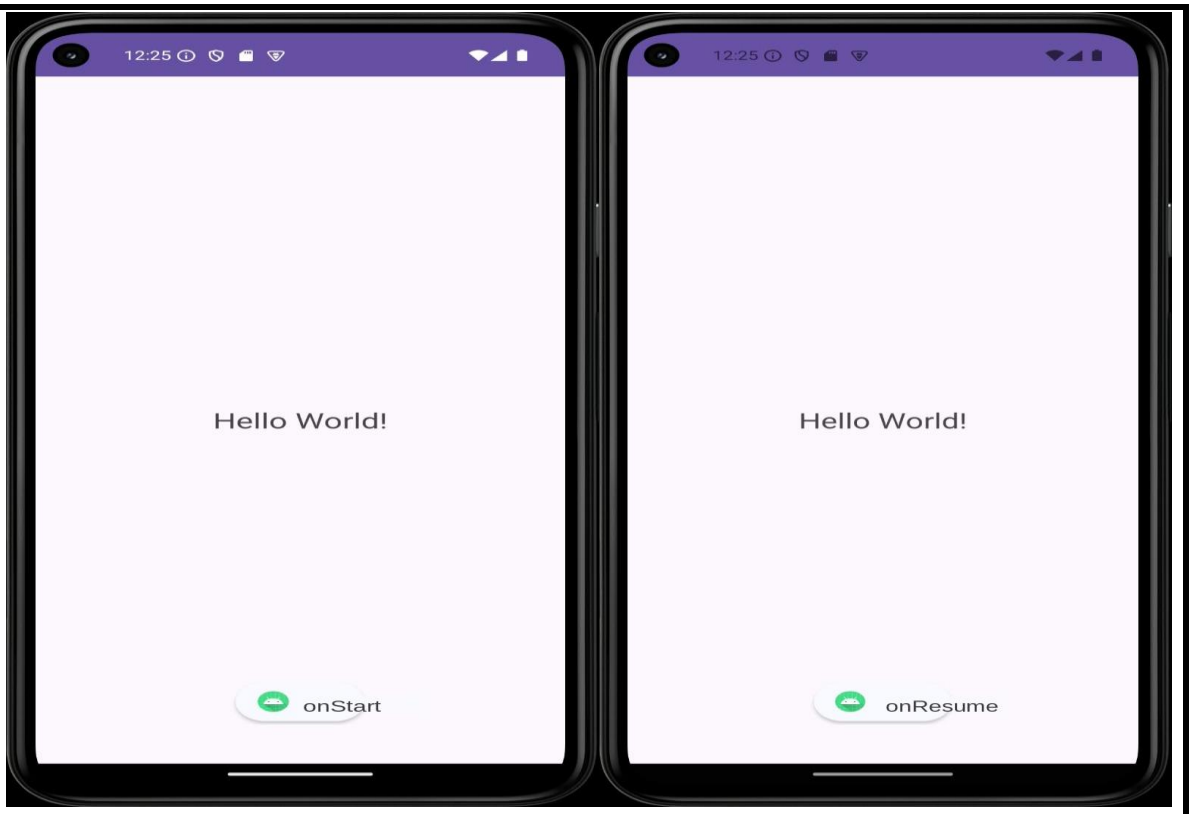
```
package com.example.program2;
import android.os.Bundle;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        Toast.makeText(this, "onCreate()", Toast.LENGTH_SHORT).show();
    }
    @Override
    protected void onStart() {
        super.onStart();
        Toast.makeText(this, "onStart()", Toast.LENGTH_SHORT).show();
    }
    @Override
    protected void onResume() {
        super.onResume();
        Toast.makeText(this, "onResume()", Toast.LENGTH_SHORT).show();
    }
    @Override
    protected void onPause() {
        super.onPause();
        Toast.makeText(this, "onPause()", Toast.LENGTH_SHORT).show();
    }
```

```
    @Override
    protected void onStop() {
        super.onStop();
        Toast.makeText(this, "onStop()", Toast.LENGTH_SHORT).show();
    }
    @Override
    protected void onRestart() {
        super.onRestart();
        Toast.makeText(this, "onRestart()", Toast.LENGTH_SHORT).show();
    } }
```

activity.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output





RESULT :

The mobile application user interface was successfully designed and developed using basic building blocks and UI components. The application structure was implemented and tested on the Emulator, and the expected output screens were obtained.

PROGRAM NO:3

DATE: 14/09/26

BASIC CALCULATOR

AIM: Implementing basic arithmetic operations of a simple calculator.

MainActivity.java

```
package com.example.calculator;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
EditText number1, number2;
TextView result;
Button add, subtract, multiply, divide;
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
number1 = findViewById(R.id.editTextText);
number2 = findViewById(R.id.editTextText2);
result = findViewById(R.id.textView);
add = findViewById(R.id.button);
subtract = findViewById(R.id.button2);
multiply = findViewById(R.id.button3);
divide = findViewById(R.id.button4);
add.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v) {
```

```
if (number1.getText().toString().isEmpty() ||
number2.getText().toString().isEmpty()) {
Toast.makeText(MainActivity.this,"Enter both numbers",
Toast.LENGTH_SHORT).show();
return; }
double n1 = Double.parseDouble(number1.getText().toString());
double n2 = Double.parseDouble(number2.getText().toString());
double ans = n1 + n2;
result.setText("Result: " + ans);
}});
subtract.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v) {
if (number1.getText().toString().isEmpty() ||
number2.getText().toString().isEmpty()) {
Toast.makeText(MainActivity.this,"Enter both numbers",
Toast.LENGTH_SHORT).show();
return;}
double n1 = Double.parseDouble(number1.getText().toString());
double n2 = Double.parseDouble(number2.getText().toString());
double ans = n1 - n2;
result.setText("Result: " + ans);}});
multiply.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v) {
if (number1.getText().toString().isEmpty() ||
number2.getText().toString().isEmpty()) {
Toast.makeText(MainActivity.this,"Enter both numbers",
Toast.LENGTH_SHORT).show();
return;
}
```

```
double n1 = Double.parseDouble(number1.getText().toString());
double n2 = Double.parseDouble(number2.getText().toString());
double ans = n1 * n2;
result.setText("Result: " + ans);}});
divide.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v) {
if (number1.getText().toString().isEmpty() ||
number2.getText().toString().isEmpty()) {
Toast.makeText(MainActivity.this,"Enter both numbers",
Toast.LENGTH_SHORT).show();
return;
}
double n1 = Double.parseDouble(number1.getText().toString());
double n2 = Double.parseDouble(number2.getText().toString());
if (n2 == 0) {
Toast.makeText(MainActivity.this,"Cannot divide by zero",
Toast.LENGTH_SHORT).show();
return;
}
double ans = n1 / n2;
result.setText("Result: " + ans);
}});}
```

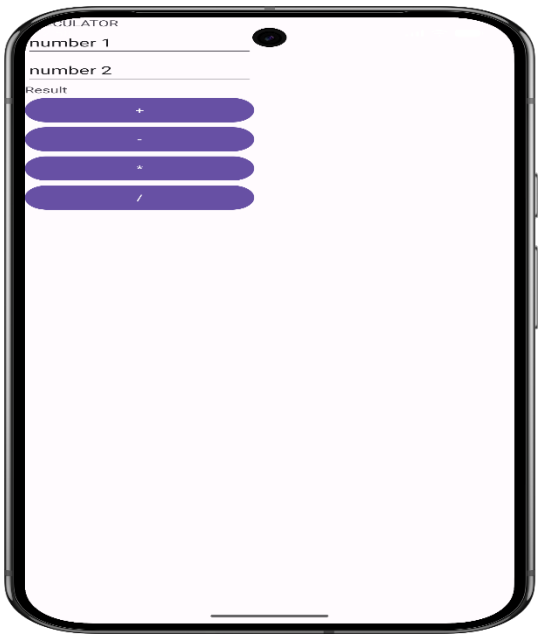
activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
```

```
android:orientation="vertical"
tools:context=".MainActivity">
<LinearLayout
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:orientation="vertical">
<TextView
android:id="@+id/textView2"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="CALCULATOR" />
<EditText
android:id="@+id/editTextText"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="text"
android:text="number 1" />
<EditText
android:id="@+id/editTextText2"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="text"
android:text="number 2" />
<TextView
android:id="@+id/textView"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Result" />
<Button
```

```
android:id="@+id/button"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="+" />
<Button
android:id="@+id/button2"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="-" />
<Button
android:id="@+id/button3"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="*" />
<Button
android:id="@+id/button4"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="/" />
</LinearLayout></LinearLayout>
```

OUTPUT





RESULT :

The mobile application user interface was successfully designed and developed using basic building blocks and UI components. The application structure was implemented and tested on the Emulator, and the expected output screens were obtained.

PROGRAM NO:4

DATE: 14/09/26

VALIDATION UI

AIM: Implement validations on various UI

MainActivity.java

```
import android.os.Bundle;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.EditText;
import android.widget.RadioGroup;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
   EditText name, email, password;
   RadioGroup genderGroup;
   CheckBox agree;
   Button submit;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      name = findViewById(R.id.name);
      email = findViewById(R.id.email);
      password = findViewById(R.id.password);
      genderGroup = findViewById(R.id.genderGroup);
      agree = findViewById(R.id.agree);
      submit = findViewById(R.id.submit);
      submit.setOnClickListener(v -> {
         String n = name.getText().toString().trim();
         String e = email.getText().toString().trim();
```

```
String p = password.getText().toString().trim();
if (n.isEmpty()) {
    name.setError("Enter your name");
    name.requestFocus();
}
else if (e.isEmpty()) {
    email.setError("Enter your email");
    email.requestFocus();
}
else if (!e.contains("@")) {
    email.setError("Enter a valid email");
    email.requestFocus();
}
else if (p.isEmpty()) {
    password.setError("Enter password");
    password.requestFocus();
}
else if (p.length() < 6) {
    password.setError("Password must be at least 6 characters");
    password.requestFocus();
}
else if (genderGroup.getCheckedRadioButtonId() == -1) {
    Toast.makeText(this,
            "Select your gender",
            Toast.LENGTH_SHORT).show();
}
else if (!agree.isChecked()) {
    Toast.makeText(this,
            "Please agree to the terms and conditions",
            Toast.LENGTH_SHORT).show();
}
else {
```

```
            Toast.makeText(this,
                    "Registration Successful",
                    Toast.LENGTH_LONG).show();
        }
    });
  }
}
```

activity.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="24dp">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Registration Form"
      android:textSize="26sp"
      android:textStyle="bold"
      android:layout_gravity="center"
      android:layout_marginBottom="25dp" />

   <!-- Name -->
   <EditText
      android:id="@+id/name"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Name"
      android:inputType="textPersonName"
      android:layout_marginBottom="12dp" />
```

```
<!-- Email -->
<EditText
    android:id="@+id/email"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Email"
    android:inputType="textEmailAddress"
    android:layout_marginBottom="12dp" />

<!-- Password -->
<EditText
    android:id="@+id/password"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Password"
    android:inputType="textPassword"
    android:layout_marginBottom="15dp" />

<!-- Gender -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Gender"
    android:textSize="18sp"
    android:textStyle="bold" />

<RadioGroup
    android:id="@+id/genderGroup"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
```

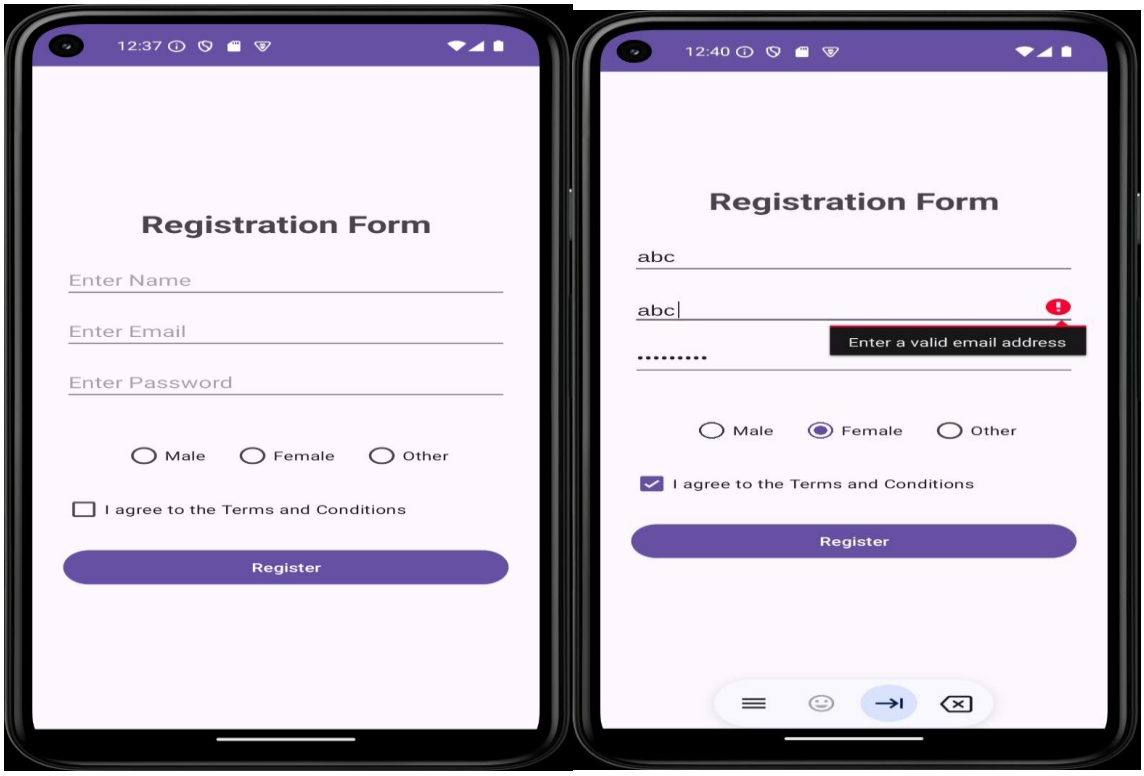
```
        android:layout_marginBottom="15dp">

        <RadioButton
            android:id="@+id/male"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Male" />

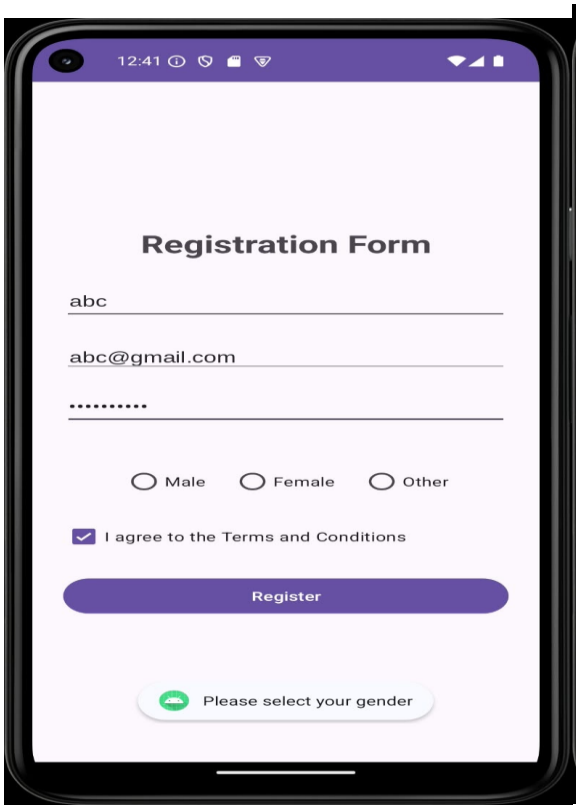
        <RadioButton
            android:id="@+id/female"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Female"
            android:layout_marginStart="30dp" />
    </RadioGroup>
    <!-- Agree Checkbox -->
    <CheckBox
        android:id="@+id/agree"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="I agree to the terms and conditions"
        android:layout_marginBottom="20dp" />
    <!-- Submit -->
    <Button
        android:id="@+id/submit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="SUBMIT" />

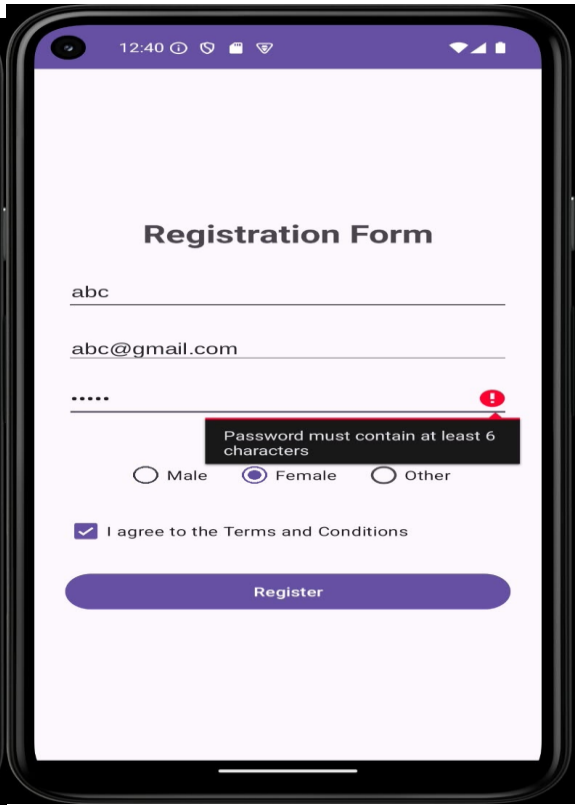
</LinearLayout>
```

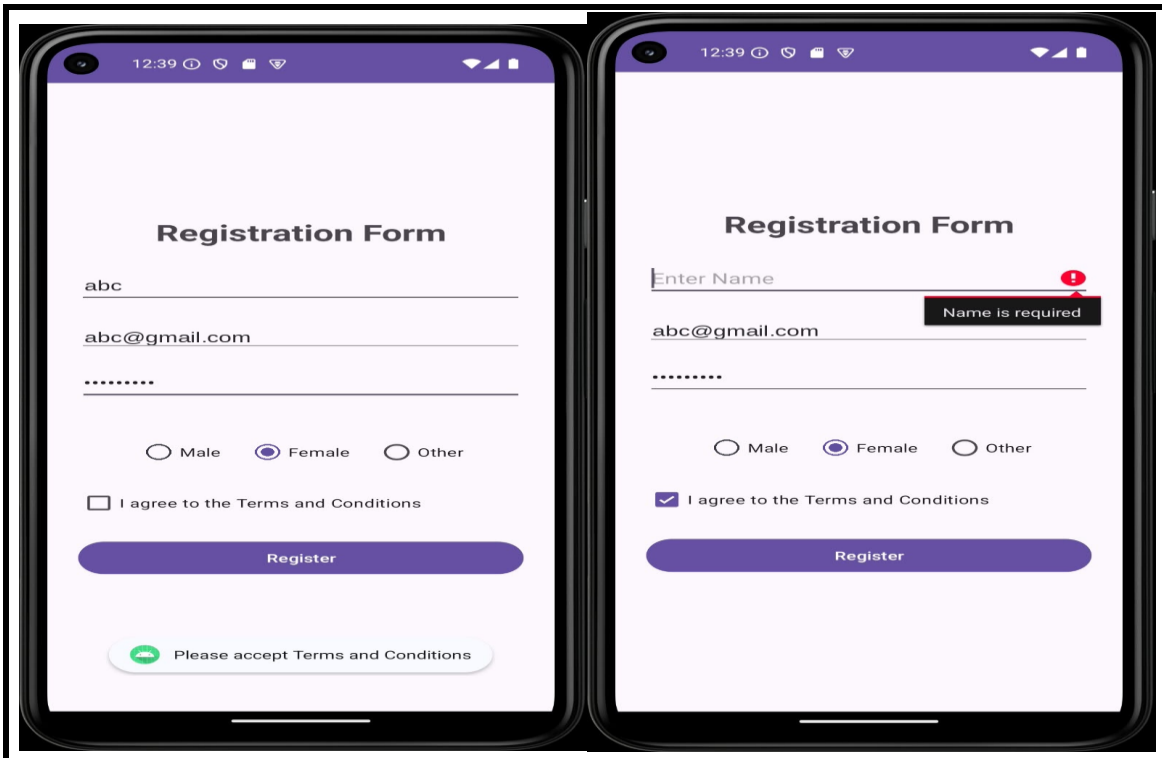
OUTPUT

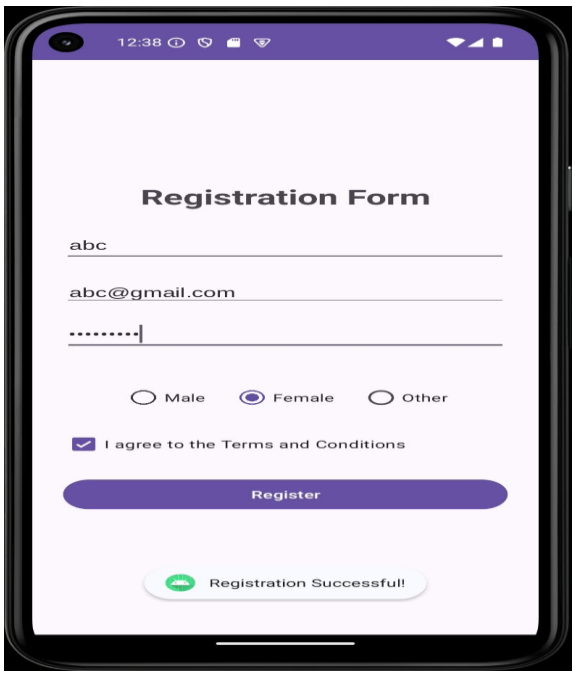

12:37
Registration Form
Enter Name
Enter Email
Enter Password
Male
Female
Other
I agree to the Terms and Conditions
Register

12:40
Registration Form
abc
abc
Enter a valid email address
Male
Female
Other
I agree to the Terms and Conditions
Register


12:41
Registration Form
abc
abc@gmail.com
Male
Female
Other
I agree to the Terms and Conditions
Register
Please select your gender


12:40
Registration Form
abc
abc@gmail.com
Password must contain at least 6 characters
Male
Female
Other
I agree to the Terms and Conditions
Register





RESULT :

The mobile application user interface was successfully designed and developed using basic building blocks and UI components. The application structure was implemented and tested on the Emulator, and the expected output screens were obtained.

PROGRAM NO:5

DATE:

SIMPLE CALCULATOR USING GRID LAYOUT

AIM: Design a simple Calculator using Grid Layout and Cascaded Linear Layout.

MainActivity.java

```
package com.example.program5;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
public class MainActivity extends AppCompatActivity implements View.OnClickListener {
TextView number;
double num1 = 0;
double num2 = 0;
String op = "";
Button b0, b1, b2, b3, b4, b5, b6, b7, b8, b9;
Button sum, sub, mul, div, equal, clear;
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
number = findViewById(R.id.et_number);
b0 = findViewById(R.id.b0);
b1 = findViewById(R.id.b1);
b2 = findViewById(R.id.b2);
b3 = findViewById(R.id.b3);
b4 = findViewById(R.id.b4);
b5 = findViewById(R.id.b5);
b6 = findViewById(R.id.b6);
b7 = findViewById(R.id.b7);
b8 = findViewById(R.id.b8);
b9 = findViewById(R.id.b9);
sum = findViewById(R.id.add);
sub = findViewById(R.id.subtract);
mul = findViewById(R.id.multiply);
div = findViewById(R.id.divide);
equal = findViewById(R.id.equal);
clear = findViewById(R.id.delete);
b0.setOnClickListener(this);
b1.setOnClickListener(this);
b2.setOnClickListener(this);
b3.setOnClickListener(this);
b4.setOnClickListener(this);
b5.setOnClickListener(this);
```

```
b6.setOnClickListener(this);
b7.setOnClickListener(this);
b8.setOnClickListener(this);
b9.setOnClickListener(this);
sum.setOnClickListener(this);
sub.setOnClickListener(this);
mul.setOnClickListener(this);
div.setOnClickListener(this);
equal.setOnClickListener(this);
clear.setOnClickListener(this);
}
@Override
public void onClick(View view) {
Button b = (Button) view;
// Clear button
if (view == clear) {
number.setText("");
num1 = 0;
num2 = 0;
op = "";
}
// Operator buttons
else if (view == sum || view == sub ||
view == mul || view == div) {
if (number.getText().toString().isEmpty()) {
return;
}
num1 = Double.parseDouble(number.getText().toString());
op = b.getText().toString();
number.setText("");
}
// Equal button
else if (view == equal) {
if (number.getText().toString().isEmpty() || op.isEmpty()) {
return;
}
num2 = Double.parseDouble(number.getText().toString());
double result = 0;
if (op.equals("+")) {
result = num1 + num2;
}
else if (op.equals("-")) {
result = num1 - num2;
}
else if (op.equals("×")) {
result = num1 * num2;
}
else if (op.equals("/")) {
if (num2 == 0) {
number.setText("Error");
```

```
return;
}
result = num1 / num2;
}
number.setText(String.valueOf(result));
op = "";
}
// Number buttons
else {
String s = number.getText().toString();
number.setText(s + b.getText().toString());
}
}
}
```

activity_main.xml

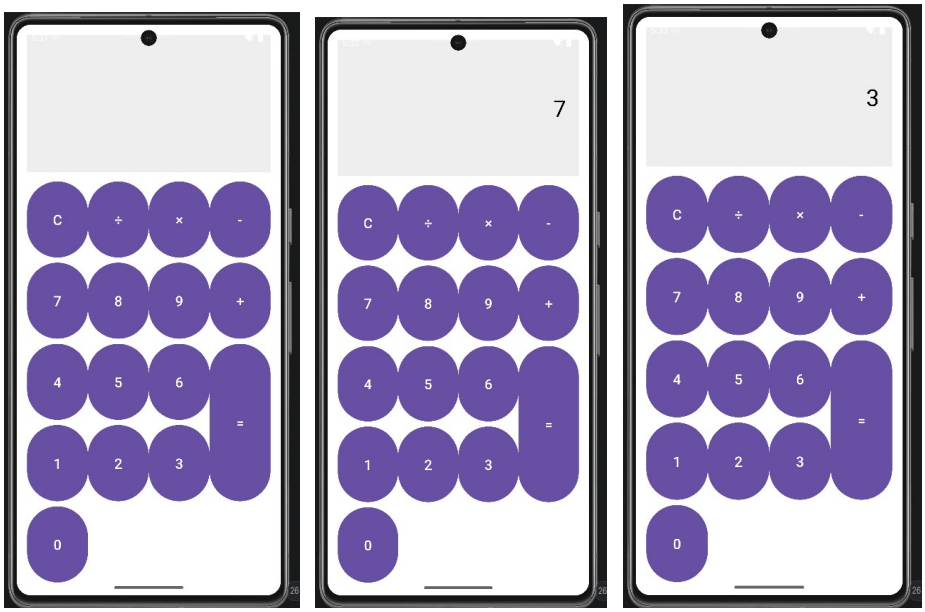
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="16dp"
android:background="#FFFFFF">
<!-- Calculator Display -->
<TextView
android:id="@+id/et_number"
android:layout_width="match_parent"
android:layout_height="0dp"
android:layout_weight="1"
android:gravity="center_vertical|right"
android:padding="20dp"
android:text=""
android:textColor="#000000"
android:textSize="36sp"
android:background="#EEEEEE" />
<!-- Calculator Buttons -->
<GridLayout
android:layout_width="match_parent"
android:layout_height="0dp"
android:layout_weight="3"
android:columnCount="4"
android:rowCount="5"
android:paddingTop="10dp">
<!-- Row 1 -->
<Button
android:id="@+id/delete"
android:layout_width="0dp"
android:layout_height="0dp"
android:layout_columnWeight="1"
```

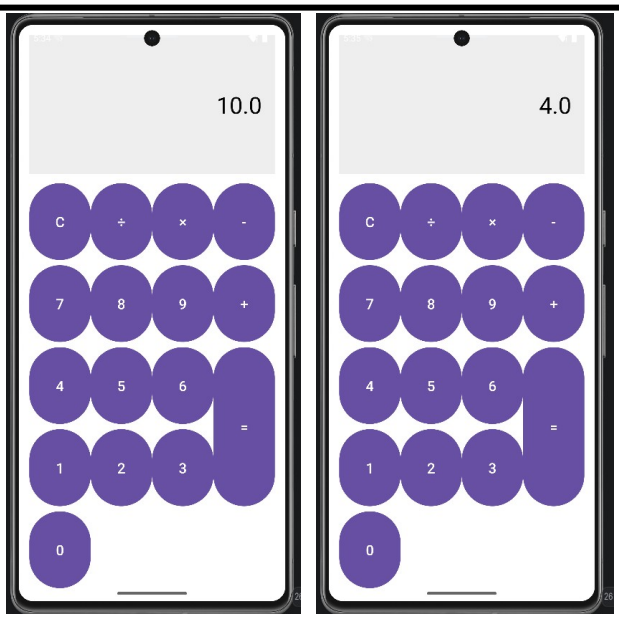
```
android:layout_rowWeight="1"
android:text="C"
android:textSize="22sp" />
<Button
android:id="@+id/divide"
android:layout_width="0dp"
android:layout_height="0dp"
android:layout_columnWeight="1"
android:layout_rowWeight="1"
android:text="/"
android:textSize="22sp" />
<Button
android:id="@+id/multiply"
android:layout_width="0dp"
android:layout_height="0dp"
android:layout_columnWeight="1"
android:layout_rowWeight="1"
android:text="×"
android:textSize="22sp" />
<Button
android:id="@+id/subtract"
android:layout_width="0dp"
android:layout_height="0dp"
android:layout_columnWeight="1"
android:layout_rowWeight="1"
android:text="-"
android:textSize="22sp" />
<!-- Row 2 -->
<Button
android:id="@+id/b7"
android:layout_width="0dp"
android:layout_height="0dp"
android:layout_columnWeight="1"
android:layout_rowWeight="1"
android:text="7"
android:textSize="22sp" />
<Button
android:id="@+id/b8"
android:layout_width="0dp"
android:layout_height="0dp"
android:layout_columnWeight="1"
android:layout_rowWeight="1"
android:text="8"
android:textSize="22sp" />
<Button
android:id="@+id/b9"
android:layout_width="0dp"
android:layout_height="0dp"
android:layout_columnWeight="1"
android:layout_rowWeight="1"
```

```
android:text="9"
android:textSize="22sp" />
<Button
android:id="@+id/add"
android:layout_width="0dp"
android:layout_height="0dp"
android:layout_columnWeight="1"
android:layout_rowWeight="1"
android:text="+"
android:textSize="22sp" />
<!-- Row 3 -->
<Button
android:id="@+id/b4"
android:layout_width="0dp"
android:layout_height="0dp"
android:layout_columnWeight="1"
android:layout_rowWeight="1"
android:text="4"
android:textSize="22sp" />
<Button
android:id="@+id/b5"
android:layout_width="0dp"
android:layout_height="0dp"
android:layout_columnWeight="1"
android:layout_rowWeight="1"
android:text="5"
android:textSize="22sp" />
<Button
android:id="@+id/b6"
android:layout_width="0dp"
android:layout_height="0dp"
android:layout_columnWeight="1"
android:layout_rowWeight="1"
android:text="6"
android:textSize="22sp" />
<Button
android:id="@+id/equal"
android:layout_width="0dp"
android:layout_height="0dp"
android:layout_columnWeight="1"
android:layout_rowWeight="2"
android:layout_rowSpan="2"
android:text="="
android:textSize="22sp" />
<!-- Row 4 -->
<Button
android:id="@+id/b1"
android:layout_width="0dp"
android:layout_height="0dp"
android:layout_columnWeight="1"
```

```
android:layout_rowWeight="1"
android:text="1"
android:textSize="22sp" />
<Button
android:id="@+id/b2"
android:layout_width="0dp"
android:layout_height="0dp"
android:layout_columnWeight="1"
android:layout_rowWeight="1"
android:text="2"
android:textSize="22sp" />
<Button
android:id="@+id/b3"
android:layout_width="0dp"
android:layout_height="0dp"
android:layout_columnWeight="1"
android:layout_rowWeight="1"
android:text="3"
android:textSize="22sp" />
<!-- Row 5 -->
<Button
android:id="@+id/b0"
android:layout_width="0dp"
android:layout_height="0dp"
android:layout_columnWeight="1"
android:layout_rowWeight="1"
android:text="0"
android:textSize="22sp" />
</GridLayout>
</LinearLayout>
```

Output





RESULT :

The simple calculator was successfully designed using Grid Layout and Linear Layout. The arithmetic operations were performed correctly, and the application was tested on the Emulator and the expected output screens were obtained.

PROGRAM NO:6

DATE:

REGISTRATION FORM

AIM: Design a registration activity and store registration details in local memory of the phone using Intents and Shared Preferences.

MainActivity.java

```
package com.example.pgrm6sharedpref;
import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.EditText;
import android.widget.RadioGroup;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
    EditText name, email, password;
    RadioGroup genderGroup;
    CheckBox agree;
    Button submit;
    SharedPreferences preferences;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        name = findViewById(R.id.name);
        email = findViewById(R.id.email);
        password = findViewById(R.id.password);
        genderGroup = findViewById(R.id.genderGroup);
        agree = findViewById(R.id.agree);
        submit = findViewById(R.id.submit);
        preferences = getSharedPreferences(
                "RegistrationData",
                MODE_PRIVATE
        );
        submit.setOnClickListener(v -> {
            String n = name.getText().toString().trim();
            String e = email.getText().toString().trim();
            String p = password.getText().toString().trim();
            if (n.isEmpty()) {
                name.setError("Enter your name");
                name.requestFocus();
                return;
            }
            if (e.isEmpty()) {
```

```
            email.setError("Enter your email");
            email.requestFocus();
            return;
        }
        if (!e.contains("@")) {
            email.setError("Enter a valid email");
            email.requestFocus();
            return;
        }
        if (p.isEmpty()) {
            password.setError("Enter password");
            password.requestFocus();
            return;
        }
        if (p.length() < 6) {
            password.setError(
                    "Password must be at least 6 characters"
            );
            password.requestFocus();
            return;
        }
        int genderId = genderGroup.getCheckedRadioButtonId();
        if (genderId == -1) {
            Toast.makeText(
                    this,
                    "Select your gender",
                    Toast.LENGTH_SHORT
            ).show();
            return;
        }
        if (!agree.isChecked()) {
            Toast.makeText(
                    this,
                    "Please agree to the terms and conditions",
                    Toast.LENGTH_SHORT
            ).show();
            return;
        }
        String gender;
        if (genderId == R.id.male) {
            gender = "Male";
        } else {
            gender = "Female";
        }
        // Store data using SharedPreferences
        SharedPreferences.Editor editor =
                preferences.edit();
        editor.putString("name", n);
        editor.putString("email", e);
        editor.putString("password", p);
```

```
            editor.putString("gender", gender);
            editor.apply();
            Toast.makeText(
                    this,
                    "Registration Successful",
                    Toast.LENGTH_SHORT
            ).show();
            // Navigate to second activity
            Intent intent = new Intent(MainActivity.this, MainActivity2.class);
            startActivity(intent);
            finish();
        });
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Registration Form"
        android:textSize="26sp"
        android:textStyle="bold"
        android:layout_gravity="center"
        android:layout_marginBottom="25dp" />
    <EditText
        android:id="@+id/name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name"
        android:inputType="textPersonName"
        android:layout_marginBottom="12dp" />
    <EditText
        android:id="@+id/email"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Email"
        android:inputType="textEmailAddress"
        android:layout_marginBottom="12dp" />
    <EditText
        android:id="@+id/password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Password"
```

```
    android:inputType="textPassword"
    android:layout_marginBottom="15dp" />
  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Gender"
    android:textSize="18sp"
    android:textStyle="bold" />
  <RadioGroup
    android:id="@+id/genderGroup"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginBottom="15dp">
    <RadioButton
      android:id="@+id/male"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Male" />
    <RadioButton
      android:id="@+id/female"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Female"
      android:layout_marginStart="30dp" />
  </RadioGroup>
  <CheckBox
    android:id="@+id/agree"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="I agree to the terms and conditions"
    android:layout_marginBottom="20dp" />
  <Button
    android:id="@+id/submit"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="SUBMIT" />
</LinearLayout>
```

MainActivity2.java

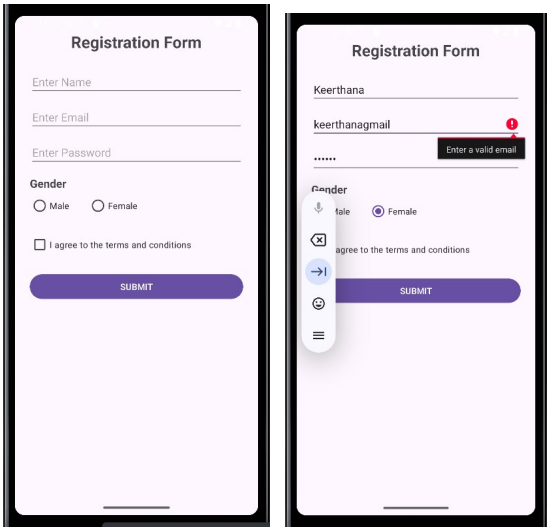
```
package com.example.pgrm6sharedpref;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class MainActivity2 extends AppCompatActivity {
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main2);
  }
```

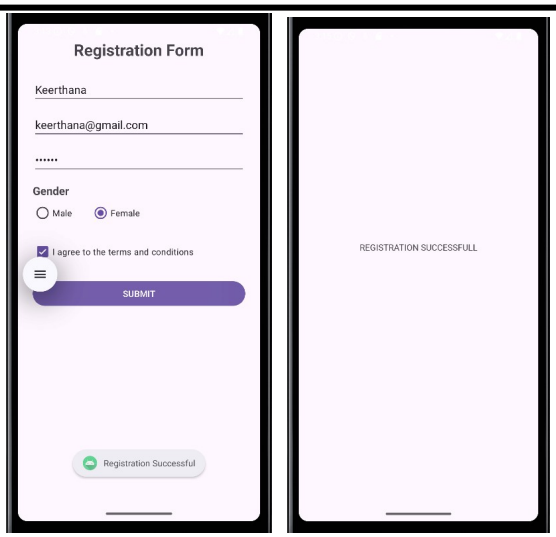
}

activity_main2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity2">
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical">
        <TextView
            android:id="@+id/textView4"
            android:layout_width="match_parent"
            android:layout_height="716dp"
            android:gravity="center"
            android:text="REGISTRATION SUCCESSFULL" />
    </LinearLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output





RESULT :

The registration activity was successfully designed. The registration details were validated and stored in the local memory of the phone using Shared Preferences, and the second activity was opened using an Intent. The expected output screens were obtained on the Emulator.

PROGRAM NO:7

DATE:

FACEBOOK PAGE

AIM: Create a Facebook page using Relative Layout and set the properties using the .xml file.

MainActivity.java

```
package com.example.program7;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
    private EditText e_name;
    private EditText e_pwd;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        e_name = findViewById(R.id.editTextText2);
        e_pwd = findViewById(R.id.editTextPassword);
    }
    public void onclick(View view) {
        String username = e_name.getText().toString();
        String password = e_pwd.getText().toString();
        if(username.equals("mca") && password.equals("mca123"))
        {
            Toast.makeText(this, "login successfully", Toast.LENGTH_SHORT).show();
        } else {
            Toast.makeText(this, "invalid username", Toast.LENGTH_SHORT).show();
        }
    }
}
```

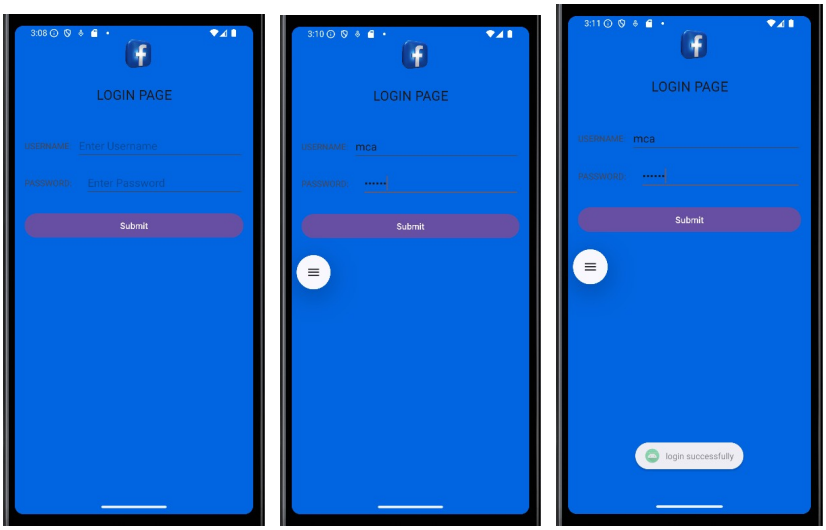
activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/ed_id"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:background="#0165E1"
```

```
tools:context=".MainActivity">
<RelativeLayout
   android:layout_width="375dp"
   android:layout_height="59dp">
   <ImageView
      android:id="@+id/imageView3"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      app:srcCompat="@drawable/facebook" />
</RelativeLayout>
<TextView
   android:id="@+id/textView4"
   android:layout_width="match_parent"
   android:layout_height="101dp"
   android:gravity="center"
   android:paddingBottom="24dp"
   android:text="LOGIN PAGE"
   android:textAppearance="@style/TextAppearance.AppCompat.Large" />
<LinearLayout
   android:layout_width="match_parent"
   android:layout_height="wrap_content"
   android:layout_marginBottom="16dp"
   android:orientation="horizontal">
   <TextView
      android:id="@+id/textView3"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_gravity="center_vertical"
      android:layout_marginRight="8dp"
      android:text="USERNAME:" />
   <EditText
      android:id="@+id/editTextText2"
      android:layout_width="0dp"
      android:layout_height="wrap_content"
      android:layout_weight="1"
      android:ems="10"
      android:hint="Enter Username"
      android:inputType="text" />
</LinearLayout>
<LinearLayout
   android:layout_width="match_parent"
   android:layout_height="wrap_content"
   android:layout_marginBottom="24dp"
   android:orientation="horizontal">
   <TextView
      android:id="@+id/textViewPassword"
      android:layout_width="100dp"
      android:layout_height="wrap_content"
      android:layout_gravity="center_vertical"
      android:text="PASSWORD:" />
```

```
        <EditText
            android:id="@+id/editTextPassword"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:ems="10"
            android:hint="Enter Password"
            android:inputType="textPassword" />
    </LinearLayout>
    <Button
        android:id="@+id/button2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:onClick="onclick"
        android:text="Submit" />
</LinearLayout>
```

Output



RESULT :

The Facebook login page was successfully created using Relative Layout and its properties were set using the .xml file. The application was tested on the Emulator and the expected output screens were obtained.

PROGRAM NO:8

DATE:

TOGGLE IMAGE USING FRAME LAYOUT

AIM: Develop an application that toggles an image using Frame Layout.

MainActivity.java

```
package com.example.program8;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.ImageView;
public class MainActivity extends AppCompatActivity implements View.OnClickListener{
   ImageView img,img1;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      img=findViewById(R.id.imageView4);
      img1=findViewById(R.id.imageView3);
      img.setOnClickListener(this);
      img1.setOnClickListener(this);
   }
   @Override
   public void onClick(View v){
      if(v==img){
         img.setVisibility(View.GONE);
         img1.setVisibility(View.VISIBLE);
      }else{
         img1.setVisibility(View.GONE);
         img.setVisibility(View.VISIBLE);
      }
   }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center">
   <ImageView
      android:id="@+id/imageView4"
      android:layout_width="300dp"
      android:layout_height="300dp"
      android:src="@drawable/im1"
      android:scaleType="centerCrop"
```

```
        android:contentDescription="Image 1" />
    <ImageView
        android:id="@+id/imageView3"
        android:layout_width="300dp"
        android:layout_height="300dp"
        android:src="@drawable/im2"
        android:scaleType="centerCrop"
        android:contentDescription="Image 2"
        android:visibility="gone" />
</LinearLayout>
```

Output





RESULT :

The application that toggles an image on click was successfully developed using Frame Layout. It was tested on the Emulator and the expected output screens were obtained.

PROGRAM NO:9

DATE:

INTENT

AIM: Implement Intent to navigate between multiple activities.

MainActivity.java

```
package com.example.program9;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
   EditText name,classs,rollno;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      name = findViewById(R.id.edit1);
      classs = findViewById(R.id.editTextText);
      rollno = findViewById(R.id.editTextText2);
   }
   public void onClick(View v) {
      String val_1 = name.getText().toString();
      String val_2= classs.getText().toString();
      String val_3 = rollno.getText().toString();
      Intent i = new Intent(MainActivity.this, MainActivity2.class);
      i.putExtra("name", val_1);
      i.putExtra("class", val_2);
      i.putExtra("roll no", val_3);
      startActivity(i);
   }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">
   <LinearLayout
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:orientation="vertical">
```

```
    <EditText
        android:id="@+id/edit1"
        android:layout_width="match_parent"
        android:layout_height="66dp"
        android:ems="10"
        android:inputType="text"
        android:text="NAME" />
    <EditText
        android:id="@+id/editTextText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="text"
        android:text="CLASS" />
    <EditText
        android:id="@+id/editTextText2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="text"
        android:text="ROLLNO" />
    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:onClick="onClick"
        android:text="Button" />
  </LinearLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity2.java

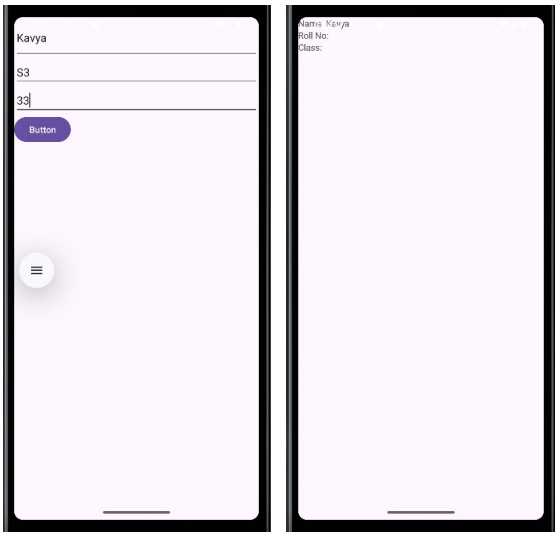
```
package com.example.program9;
import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity2 extends AppCompatActivity {
  TextView textView, textView2, textView3;
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main2);
    textView = findViewById(R.id.textView);
    textView2 = findViewById(R.id.textView2);
    textView3 = findViewById(R.id.textView3);
    // Receive name from MainActivity
    String name = getIntent().getStringExtra("name");
    textView.setText("Name: " + name);
    // For now, display Roll No and Class
    textView2.setText("Roll No: ");
```

```
        textView3.setText("Class: ");
    }
}
```

activity_main2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity2">
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical">
        <TextView
            android:id="@+id/textView"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="TextView" />
        <TextView
            android:id="@+id/textView2"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="TextView" />
        <TextView
            android:id="@+id/textView3"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="TextView" />
    </LinearLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output



RESULT :

Intent was successfully implemented to navigate between multiple activities and pass data from one activity to another. The application was tested on the Emulator and the expected output screens were obtained.

PROGRAM NO:10

DATE:

OPTIONS MENU

AIM: Implement Options Menu to navigate to activities.

MainActivity.java

```
package com.example.program10;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuInflater;
import android.view.MenuItem;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        MenuInflater inflater=getMenuInflater();
        inflater.inflate(R.menu.samplemenu,menu);
        return super.onCreateOptionsMenu(menu);
    }
    public boolean onOptionsItemSelected (MenuItem item){
        int itemId = item.getItemId();
        if (itemId == R.id.item1) {
            Toast.makeText(this, "Item 1 Selected", Toast.LENGTH_SHORT).show();
        } else if (itemId == R.id.item2) {
            Intent i=new Intent(this,MainActivity2.class);
            startActivity(i);
        }
        return true;
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
```

```
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

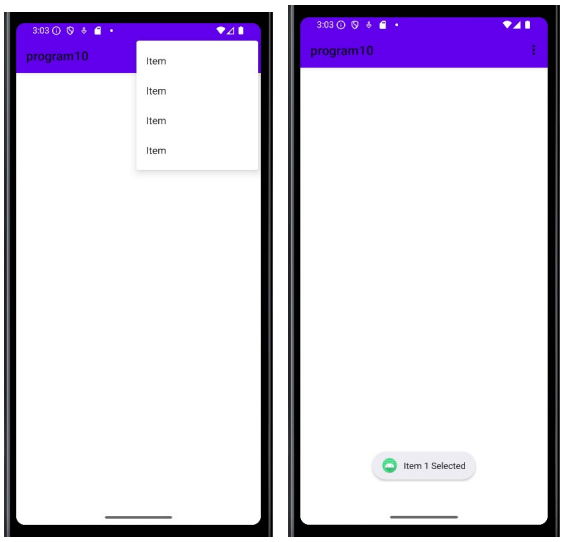
MainActivity2.java

```
package com.example.program10;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Toast;
public class MainActivity2 extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
        Toast.makeText(this, "Item 2 Selected", Toast.LENGTH_SHORT).show();
    }
}
```

activity_main2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity2">
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output



RESULT :

The Options Menu was successfully implemented to display a message and to navigate to another activity. The application was tested on the Emulator and the expected output screens were obtained.

PROGRAM NO:11

DATE:

ARRAYADAPTER WITH LISTVIEW

AIM: Develop an application that uses ArrayAdapter with ListView.

MainActivity.java

```
package com.example.program11;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class MainActivity extends AppCompatActivity {
    ListView lv;
    String[] s={"Jan","feb","Mar","Apr","May"};
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        lv=findViewById(R.id.listid);
        ArrayAdapter<String>adapter=new
ArrayAdapter<String>(this,R.layout.layout,R.id.MONTH,s);
        lv.setAdapter(adapter);
        lv.setOnItemClickListener(new AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> adapterView, View view, int i, long l) {

Toast.makeText(getApplicationContext(),lv.getItemAtPosition(i).toString(),Toast.LENGTH_
LONG).show();
            }
        });
    }
}
```

activity_main.xml

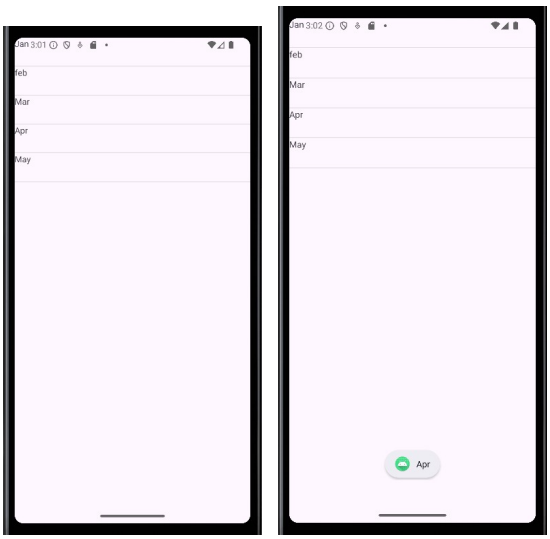
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
```

```
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <LinearLayout
        android:id="@+id/linearLayout"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical">
        <ListView
            android:id="@+id/listid"
            android:layout_width="409dp"
            android:layout_height="354dp" />
    </LinearLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```

layout.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <TextView
        android:id="@+id/MONTH"
        android:layout_width="wrap_content"
        android:layout_height="47dp"
        android:layout_weight="1"
        android:text="Month" />
</LinearLayout>
```

Output



RESULT :

The application that displays a list of items using ArrayAdapter with ListView was successfully developed. The selected item is displayed as a Toast message. It was tested on the Emulator and the expected output screens were obtained.